Quiz Master - V1

Author Information

- **Name:** Prashant Ranjan
- **Roll No:** 23f2002267
- **Email:** 23f2002267@ds.study.iitm.ac.in

About Me

I'm pursuing my **BS in Data Science** at **IIT Madras**. Through this **Quiz Master** project, I have gained valuable experience in web development using technologies like **Python**, **Flask**, **SQLAlchemy**, **HTML** & **CSS**.

Project Description

Quiz Master - V1 is a multi-user web application designed for exam preparation. The platform provides two distinct roles:

- **Admin (Quiz Master)**: Manages subjects, chapters, and quizzes.
- **Users**: Can register, attempt quizzes, and review their scores.

Technology Stack

- **Python**: Backend development
- **HTML, CSS, and Bootstrap**: Front-end design
- **Flask**: Web framework for routing, rendering templates, and handling requests
- **Flask-SQLAlchemy**: ORM for database management

- **Jinja2**: Template engine for dynamic HTML rendering
- **SQLite**: Database engine for persistent data storage

Roles and Responsibilities

Admin (Quiz Master)

- Superuser with root access
- Create and manage subjects, chapters, and quizzes
- Monitor user activity and quiz performance

User

- Register and log into the platform
- Select subjects and chapters to attempt quizzes
- Review quiz results and scores

Database Schema Design

User Table

- **id** (Primary Key)
- **username** (Unique and required)
- **password_hash** (Required)
- **fullname** (Required)
- **qualification** (Required)
- **dob** (Required)

- **is_admin** (Boolean, default **False**)
- **is_active** (Boolean, default **True**)

Subject Table

- **id** (Primary Key)
- **name** (Unique and required)
- **description**

Chapter Table

- **id** (Primary Key)
- **subject_id** (Foreign Key linking to Subject)
- **name** (Required)
- **description**

Quiz Table

- **id** (Primary Key)
- **chapter_id** (Foreign Key linking to Chapter)
- **time_duration** (Required)
- **questions**
- **remarks**

Question Table

- **id** (Primary Key)
- **chapter_id** (Foreign Key linking to Chapter)
- **q_statement** (Unique and required)
- **options** (Stored as CSV text)

- **correct_option**

Score Table

- **id** (Primary Key)
- **quiz_id** (Foreign Key linking to Quiz)
- **user_id** (Foreign Key linking to User)
- **timestamp_of_attempt** (Timestamp)
- **total_score** (Required)

Architecture and Features

Quiz Master - V1 follows the **MVC (Model-View-Controller)** architecture:

- **Models**: Built using Flask-SQLAlchemy to represent database tables.
- **Views**: Developed using Jinja2 templates with HTML and CSS for an intuitive user experience.
- **Controllers**: Implemented using Flask routes to handle user requests and apply business logic.

Video Demonstration

Watch the Quiz Master Demo

Contact

For further inquiries, feel free to reach out at: 23f2002267@ds.study.iitm.ac.in